A fifth postulate for the Principia: joint determinacy of place and motion

Result, 2026-09-27; revised the same day after an adversarial Fable review. The user's goal: identify the statement of the *Principia* whose change gives mechanics with $h > 0$, as the parallel postulate does for curved geometry. The proposed statement is **joint determinacy**: every body admits states in which its place and its quantity of motion are given together, sharply, at one instant. Newton asserts it in the scholium closing Book I, Section I, as his reply to the objection that a vanishing ratio is not ultimate before the quantities vanish and is nothing once they have: the ultimate velocity is "the very velocity with which the body reaches its last place", a limit "certain and definite" (§1). A place belongs to an instant and a motion to an interval; the classical debate from Zeno's arrow onward keeps them apart, and Newton's ultimate velocity joins them.

Joint determinacy can be denied in two ways: in the algebra of observables (they cease to commute) or in the states (a restriction on which states are admissible). For the algebraic denial the analogy with Euclid's fifth postulate holds in the following precise, partial sense.

1. **Independence (Theorem A).** For every real $\hbar$, the Moyal product $*_\hbar$ satisfies Newton's second law as an identity of observables (Heisenberg form), and for Newton's integrable cases (inertia, uniform gravity, the force proportional to distance of Book I, Proposition X) every observable evolves by the classical flow. Relative to these Heisenberg-form axioms, commutativity is undecided.
2. **One constant (Theorem B, Gutt's theorem in this setting).** Every associative product on the polynomial observables that deforms the commutative one, respects complex conjugation, and is covariant under the affine symplectic group (premise (H2), stated and justified in §4) is a Moyal product with one real constant $\hbar$, of the units of action. The state route adds one constant too (Theorem B′): a covariant, noise-closed restriction on the Gaussian states is $\sqrt{\det \Sigma} \geq \zeta$, with $\zeta = \hbar/2$ in quantum mechanics.
3. **The floor (Theorem C).** For $\hbar \neq 0$, every state obeys $\Delta q \, \Delta p \geq |\hbar|/2$, and every recorded comparison of the inertial line with the constant-force parabola, whose enclosed area Newton takes to zero in Lemmas X and XI, obeys the floor of the Planck paper, Theorem 6, with $|\hbar|$ in place of $\hbar$. For $\hbar = 0$ with all states admissible, point states give zero undetermined disturbance, so there is no floor. 3b. **Complementarity (Theorem E).** For $\hbar \neq 0$, no state confines place and momentum to windows of half-widths a, b with probabilities at least $1 - \epsilon$ unless $\lambda_0(ab/|\hbar|) \geq (1 - 2\epsilon)^2$ (Landau–Pollak–Slepian): a non-Gaussian floor on the area ab. Bohr's Como statement of complementarity names the classical "union" of space-time co-ordination and causality that this denies (§6b).
4. **Similarity (Theorem D).** The action-rescaling map $(q, p) \mapsto (\lambda q, \lambda p)$ preserves the product $*_\hbar$ only for $\hbar = 0$; up to isomorphism there are two products, commutative and Moyal. This is the mechanical counterpart of Wallis's form of the fifth postulate (similar figures of every size) and of the absolute length of Lambert and Gauss, in the sense made precise in §6.

Where the analogy stops. The state route reaches the same floor without noncommutativity: GPT-6 Astra's routes note proves the same floor in a commutative theory with a Gaussian covariance restriction, with $h_* = 2\zeta$. So "floor if and only if $\hbar \neq 0$" holds only with the state-completeness premise of Theorem C(c). Noncommutativity is the unique *algebraic* denial of joint determinacy under (H1)–(H3); it is one of two denials. And (H2) is a premise about the observables, Weyl or Groenewold covariance, which the *Principia* does not supply.

The thesis that the constant enters as a consistency condition, and the classical origin of the complex exponential, are in the user's 1998 note; §7 reads Theorems A–D against it and against Connes's tangent groupoid, whose $\varepsilon = 0$ boundary is Newton's ultimate velocity.

The mathematics is established. Moyal's product (Moyal 1949, metadata); Groenewold's reduction of its bracket to the Poisson bracket for quadratic functions (Groenewold 1946, *Physica* 12, 405, metadata) and van Hove's no-go for quantizing all observables (van Hove 1951, metadata); Vey's deformations of the Poisson bracket (Vey 1975, metadata); the deformation-quantization programme (Bayen, Flato, Fronsdal, Lichnerowicz and Sternheimer 1978, I, metadata), whose authors framed quantization as a deformation in the spirit of the passage to non-Euclidean and relativistic physics (Flato 1982; Sternheimer 1998; metadata). The uniqueness in Theorem B is Gutt's theorem: the Moyal product is the unique $Sp(2n,\mathbb{R}) \ltimes \mathbb{R}^{2n}$-invariant and covariant star product (S. Gutt, *Mém. Acad. Roy. Belg. Cl. Sci.* 44:6, 1983, as cited in Duval, El Gradechi and Ovsienko 2004, passage); the associativity step is Fletcher's (Fletcher 1990, metadata); equivalence classes of invariant products are parametrized by Bertelson, Bieliavsky and Gutt (1998) (metadata). Robertson's inequality is Robertson 1929 (metadata). The contribution here is the identification of Newton's *velocitas ultima* as the statement concerned, and the reading of Theorems A–D against the parallel postulate, with the limits just stated.

1. The textual anchor

A place is predicated of an instant; a quantity of motion, of an interval. The distinction is the substance of Zeno's arrow and of Aristotle's thesis that nothing moves in the now (*Physics* VI; cited by book only, the Greek is not held in this repository). Newton addresses it in the second paragraph of the scholium closing Book I, Section I (1687 text, held in the Latin companion, added 2026-09-27 at the user's prompting):

> Objectio est, quod quantitatum evanescentium nulla sit ultima proportio; quippe quæ, antequam evanuerunt, non est ultima, ubi evanuerunt, nulla est. Sed & eodem argumento æque contendi posset nullam esse corporis ad certum locum pergentis velocitatem ultimam. [...] Per velocitatem ultimam intelligi eam, qua corpus movetur neq; antequam attingit locum ultimum & motus cessat, neq; postea, sed tunc cum attingit, id est illam ipsam velocitatem quacum corpus attingit locum ultimum & quacum motus cessat. [...] Extat limes quem velocitas in fine motus attingere potest, non autem transgredi. Hæc est velocitas ultima. [...] Cumq; hic limes sit certus & definitus, Problema est vere Geometricum eundem determinare.

"The objection is that vanishing quantities have no ultimate proportion, since before they vanish it is not ultimate and when they have vanished it is none. By the same argument one could claim that a body reaching a given place has no ultimate velocity [...]. By the ultimate velocity is meant that with which the body moves neither before it reaches its last place and the motion ceases, nor after, but when it reaches it, that is, the very velocity with which the body reaches its last place [...]. There is a limit which the velocity at the end of the motion can reach and not pass. This is the ultimate velocity. [...] And since this limit is certain and definite, determining it is a truly geometrical problem."

A velocity, a ratio over an interval, is here assigned as a certain and definite quantity to the instant of arrival at a place. The third paragraph adds that the compared quantities are "semper diminuendas sine limite", always to be diminished without limit. The scholium defends the lemmas of Section I; Newton applies those lemmas to the sagitta of Lemma X and the polygon of Proposition I, and the Scholium to the Definitions defines motion as "translatio corporis de loco [...] in locum" (location only; the Latin of that scholium is not held here). For a geometrical ratio the thesis is a theorem about limits and holds in every model below. For a *recorded* comparison, where the place is read at an instant and the motion inferred over an interval, it becomes the claim that states exist in which both are sharp. That claim is joint determinacy. What remains interpretive is the extension from the geometrical thesis to records, which Newton did not distinguish.

Earlier authors on the arrow and uncertainty. Bohm's *Quantum Theory* (Prentice-Hall, 1951, ch. 8, pp. 145–148) discusses Zeno's arrow while contrasting classical definite position and velocity with the quantum description (passage as reported from a transcription by the GPT-6 Astra bibliography search of 2026-09-27; not checked here). Landsberg, *Seeking Ultimates* (IOP, 2nd ed., 2000, p. 124) connects velocity over an interval, uncertain position and the arrow (passage as reported). Vaidman (2008) links the arrow's momentum and localization to Heisenberg's relation before turning to the quantum Zeno effect (passage as reported; DOI verified). Goyal (2026), §VI.1, ties the arrow to Bohr's complementarity of coordination and causality (passage as reported; DOI verified). Wagstaff (*Philosophy Now* 45, 2004) argues that exact position and exact momentum cannot be represented at once, since one needs an instant and the other an interval (passage). Lynds (2003) proposes an indeterminacy that enables motion and distinguishes it explicitly from the indeterminacy of h (passage as reported). The philosophical debate on instantaneous velocity is Arntzenius (2000) and Smith (2003) (abstracts). The quantum Zeno effect of Misra and Sudarshan is a different phenomenon. No author found connects Newton's *velocitas ultima* to uncertainty; the statement reports the reach of the search.

2. The family of products

Observables of one degree of freedom (the transverse coordinate of Galileo's comparison) are polynomials in q and p with complex coefficients. Write $\mu(f \otimes g) = fg$ and

$$P = \partial_q \otimes \partial_p - \partial_p \otimes \partial_q, \qquad f *_\hbar g = \mu \circ \exp\Big(\frac{i\hbar}{2} P\Big)(f \otimes g),$$

a finite sum on polynomials. For $\hbar = 0$ it is the commutative product; $q *_\hbar p - p *_\hbar q = i\hbar$. The Moyal bracket is $\{f, g\}_\hbar = (f *_\hbar g - g *_\hbar f)/(i\hbar) = \frac{2}{\hbar}\, \mu \circ \sin(\frac{\hbar}{2} P)(f \otimes g)$, with $\{f, g\}_0$ the Poisson bracket.

3. Theorem A: independence from the Laws in Heisenberg form

Theorem A. Let $H = p^2/(2M) + V(q)$ with V a polynomial and $M > 0$. For every real $\hbar$:

(a) $\{q, H\}_\hbar = p/M$ and $\{p, H\}_\hbar = -V'(q)$: the Heisenberg equations are Newton's second law, $M\ddot{q} = -V'(q)$, as an identity of observables;

(b) if $\deg V \leq 2$ (inertia $V = 0$, uniform gravity $V = -Fq$, the force proportional to distance of Book I, Proposition X, $V = kq^2/2$), then $\{f, H\}_\hbar = \{f, H\}_0$ for every f: every observable evolves by the classical flow φ_t, and each φ_t is an automorphism of $*_\hbar$.

Proof. The bracket $\{f, g\}_\hbar$ is the Poisson bracket plus terms $\hbar^{2j}\, \mu\, P^{2j+1}(f \otimes g)$, $j \geq 1$, each of which puts at least three derivatives on each factor; they vanish if one factor has degree at most two. This gives (a) because q and p have degree one, and (b) because H has degree two. The flow of a quadratic Hamiltonian is an affine symplectic map, under which $*_\hbar$ is covariant: translations commute with the constant-coefficient operator P, and a linear map of determinant one leaves P invariant. □

The independence is relative to axioms stated in Heisenberg form: the Laws as identities of observables and the classical evolution of Newton's integrable cases. Both the commutative product and every Moyal product satisfy them.

4. Theorem B: one action constant, under covariance

Consider products $f * g = \mu \circ B(f \otimes g)$ on $\mathbb{C}[q, p]$ with B a bilinear map. On polynomials every bilinear map can be written as a formal bidifferential operator $B = \sum c_{\alpha\beta}(q, p)\, \partial^\alpha \otimes \partial^\beta$, so this is no restriction. Assume:

- (H1) $*$ is associative with unit 1, and $c_{00} = 1$;
- (H2) $*$ is covariant under the affine symplectic group: for every translation and every linear map A of determinant one, $(f \circ A) * (g \circ A) = (f * g) \circ A$;
- (H3) $\overline{f * g} = \bar{g} * \bar{f}$.

What (H2) says, and what it does not follow from. By Theorem A(b), the flows of Newton's integrable cases are affine symplectic maps: free flight is a shear, uniform gravity adds translations, and the harmonic flow is a rotation in suitable units; shears and rotations generate the group of linear maps of determinant one. (H2) requires that these flows act by automorphisms of the product, that is, that quadratic observables evolve classically in the deformed theory as well. The Laws alone do not impose it. The product $f * g = \mu \circ \exp(\frac{i\hbar}{2}P + \sigma\, \partial_q \otimes \partial_q)(f \otimes g)$, $\sigma \in \mathbb{R}$, is associative and unital, satisfies (H3), is covariant under translations and shears, and satisfies Theorem A(a) verbatim; it fails only under the harmonic rotation, where the product $q\,p$ drifts by a term proportional to σk (reviewer's counterexample). So (H2) is a premise of Weyl or Groenewold covariance, physically natural (the kinematics of composing observables is the same along Newton's integrable motions) and outside the *Principia*.

Theorem B. Under (H1)–(H3), $* = *_\hbar$ for a unique real $\hbar$, of the units of action.

Proof. Translation covariance makes the coefficients $c_{\alpha\beta}$ constant. Then B is a formal power series in the components of two vectors $u = \partial^{(1)}$, $v = \partial^{(2)}$, invariant under the simultaneous action of the determinant-one linear maps; by the first fundamental theorem of invariant theory for $SL(2)$, it is a function of $\det(u, v) = P$ alone: $B = F(P)$, $F(0) = 1$. For three factors put $x = P_{12}$, $y = P_{13}$, $z = P_{23}$; the three 2×2 minors of a 2×3 matrix are algebraically independent, and associativity reads $F(x)F(y + z) = F(z)F(x + y)$ as formal series. At $z = 0$ this is $F(x)F(y) = F(x + y)$, so $F(x) = e^{\kappa x}$ (Fletcher's step). Exchanging the factors reverses P and conjugation leaves it real, so (H3) gives $\bar{\kappa} = -\kappa$: $\kappa = i\hbar/2$ with $\hbar$ real. P carries the reciprocal units of $q\,p$. □

The state route also adds exactly one constant. The second denial of joint determinacy keeps the commutative algebra and restricts the states. For Newton's quadratic cases the natural class is the Gaussian states, which quadratic flows preserve (§7 explains why the quadratic case is exact). A Gaussian state of one degree of freedom has mean $m \in \mathbb{R}^2$ and covariance $\Sigma > 0$; write $\nu(\Sigma) = \sqrt{\det \Sigma}$, which has the units of action.

Theorem B′. Let $\mathcal{G}$ be a nonempty set of nondegenerate Gaussian states that is (i) invariant under the affine symplectic group, the flows of Newton's integrable cases (as in (H2)), and (ii) closed under adding independent Gaussian noise, $\Sigma \mapsto \Sigma + K$ with $K \geq 0$, which is what forgetting or coarse-graining a record does. Then there is a unique $\zeta \geq 0$ with $\{\nu(\Sigma) : \Sigma \in \mathcal{G}\} = [\zeta, \infty)$ or (ζ, ∞): the admissible Gaussian states are exactly those with $\sqrt{\det \Sigma} \geq \zeta$ (up to the endpoint). The constant ζ has the units of action; $\zeta = 0$ is joint determinacy (states arbitrarily close to point states), and $\zeta > 0$ is the state-route denial of GPT-6 Astra's routes note, whose Theorem F gives the floor with $h_* = 2\zeta$.

Proof. By Williamson's theorem in one degree of freedom, every $\Sigma > 0$ is $S(\nu I)S^{\mathsf{T}}$ with S of determinant one and $\nu = \nu(\Sigma)$; translations move the mean. So the affine symplectic group acts transitively on the Gaussian states with a given ν, and by (i) $\mathcal{G}$ is a union of such levels, fixed by a set $N \subset (0, \infty)$ of allowed ν. By (ii), if $\nu \in N$ then $\Sigma + \kappa I \in \mathcal{G}$ for all $\kappa \geq 0$, and $\nu(\Sigma + \kappa I)$ increases continuously from ν to ∞; so N is an interval unbounded above. Set $\zeta = \inf N$. □

In quantum mechanics the Gaussian states obey the Robertson–Schrödinger bound $\det \Sigma \geq \hbar^2/4$, so $\zeta = \hbar/2$ there, and Astra's $h_* = 2\zeta$ equals $\hbar$. The two routes of denial each add exactly one action constant, the algebraic one $\hbar$ (Theorem B) and the Gaussian-state one ζ (Theorem B′). They coincide in quantum mechanics, and no further premise here makes them coincide in general.

Two products up to isomorphism. The dilation $D_\lambda(q, p) = (\lambda q, \lambda p)$ satisfies $P((f \circ D_\lambda) \otimes (g \circ D_\lambda)) = \lambda^2 (P(f \otimes g)) \circ D_\lambda$, so $f \mapsto f \circ D_\lambda$ is an isomorphism from $*_{\lambda^2\hbar}$ to $*_\hbar$, and $(q, p) \mapsto (q, -p)$ carries $*_\hbar$ to $*_{-\hbar}$. All products with $\hbar \neq 0$ are isomorphic, and distinct from

the commutative one. Geometry has three constant-curvature classes; this algebraic family has two.

5. Theorem C: the floor on the recorded comparison

A state is a linear functional ω with $\omega(1) = 1$ and $\omega(\bar{f} *_\hbar f) \geq 0$; it is regular if its GNS representation integrates to Weyl operators.

Theorem C. (a) For $\hbar \neq 0$ every state satisfies $\Delta q \, \Delta p \geq |\hbar|/2$.

(b) For $\hbar \neq 0$, consider the comparison of the inertial line with the constant-force parabola over a cell of duration τ, with fall $s = F\tau^2/(2M)$ and impulse $J = F\tau$ (the Planck paper's convention), made by any instrument coupled unitarily to the body, with regular joint states, deciding between the two hypotheses for every initial state with error probability at most $\epsilon < \frac{1}{2}$. Then the undetermined impulses $\hat{D}_j$ and displacements $\hat{X}_j$, with spreads taken as suprema over the interpolating states of the Planck paper, satisfy

$$\frac{s}{8} \sum_j \Delta(\hat{D}_j) + \frac{J}{2} \sum_j \Delta(\hat{X}_j) \;\geq\; |\hbar| \arcsin(1 - 2\epsilon) > 0.$$

(c) For $\hbar = 0$, if every probability measure on phase space is an admissible state of body and pointers (state completeness), there is no floor: a pointer prepared at $p_y = 0$ and coupled by $g\, q\, p_y$ reads q with delivered impulse $D = 0$ and displacement $X = 0$.

Proof. (a) Cauchy–Schwarz for the positive form $(f, g) \mapsto \omega(\bar{f} *_\hbar g)$ applied to $q - \omega(q)$ and $p - \omega(p)$, with $q *_\hbar p - p *_\hbar q = i\hbar$ (Robertson). (b) By the Stone–von Neumann theorem a regular representation is a multiple of the Schrödinger representation with constant $\hbar$; the multiplicity space is absorbed into the apparatus. Theorem 6 of the Planck paper is proved from the Weyl relations and Mandelstam–Tamm in the Bures angle, which is independent of the representation, under exactly these quantifiers; for $\hbar < 0$ apply the reflection $(q, p) \mapsto (q, -p)$. (c) The coupling moves the pointer by gq and kicks the body by $-gp_y = 0$. □

The floor bounds the part of the comparison that Newton's scholium says is to be diminished without limit: the recorded fall and impulse. Part (c) shows that the floor needs either noncommutativity or a restriction of states; Astra's routes note realizes the second in a commutative theory with the same constants.

6. Theorem D: similarity

Theorem D. For $\lambda \neq \pm 1$, D_λ is an automorphism of the associative product $*_\hbar$ if and only if $\hbar = 0$.

Proof. D_λ carries $*_{\lambda^2\hbar}$ to $*_\hbar$, and $*_{\lambda^2\hbar} = *_\hbar$ iff $\lambda^2\hbar = \hbar$. □

The statement concerns the product, the structure that records see. D_λ also multiplies the Poisson bracket by λ^2, so it is not an automorphism of the classical Poisson algebra either; for the dynamics it is a similarity that rescales actions, in both theories. The honest form of the analogy is therefore the one stated in the dimensional note, Theorem B: similar records at every scale exist exactly when no absolute action is available. Wallis replaced Euclid's fifth postulate by the existence of similar figures of different size; Lambert and Gauss saw that its negation brings an absolute length. Theorem D is the counterpart for the product of observables, with $\hbar$ as the absolute action.

6b. Complementarity: the negated postulate in Bohr's words and in exact form

Bohr's founding statement of complementarity names Newton's joint determinacy as the thing given up. In the Como lecture (Bohr 1928, *Nature* 121, 580, p. 580; the sentence checked as quoted in Busch and Shilladay 2006, arXiv:quant-ph/0609048, §2.2.1):

> The very nature of the quantum theory thus forces us to regard the space-time co-ordination and the claim of causality, the union of which characterizes the classical theories, as complementary but exclusive features of the description, symbolizing the idealization of observation and definition respectively.

Space-time co-ordination is the place at an instant; the claim of causality is the quantity of motion governed by the Laws and conserved by the third. Their union is the *velocitas ultima* of §1, and Bohr's pair "observation and definition" is the distinction of §1 between the recorded comparison and the geometrical thesis. So the negated fifth postulate has a name, and its author stated it as a negation of the classical union.

It also has an exact, non-Gaussian form. Write P_a for the spectral projection of $q - q_0$ on $[-a, a]$ and Q_b for that of $p - p_0$ on $[-b, b]$.

Theorem E (complementarity). Let $\hbar \neq 0$ and $c = ab/|\hbar|$. In the regular Schrödinger representation, including its multiplicities, every normal state has probabilities $\alpha^2 = \langle P_a \rangle$ and $\beta^2 = \langle Q_b \rangle$ satisfy

$$\arccos\alpha + \arccos\beta \;\geq\; \arccos\sqrt{\lambda_0(c)},$$

where $\lambda_0(c) < 1$ is the largest eigenvalue of the time- and band-limiting operator of Slepian and Pollak, increasing in c, with $\lambda_0(c) \simeq 2c/\pi$ for small c. In particular, a state (a posterior after records) that confines the place to a window of half-width a and the momentum to one of half-width b, each with probability at least $1 - \epsilon$, $\epsilon < \frac{1}{2}$, obeys

$$\lambda_0\Big(\frac{ab}{|\hbar|}\Big) \;\geq\; (1-2\epsilon)^2, \qquad \text{so} \qquad ab \;\geq\; |\hbar|\, c_*(\epsilon), \quad c_*(\epsilon) = \lambda_0^{-1}\big((1-2\epsilon)^2\big) > 0.$$

For $\epsilon \to 0$ no state reaches it at all: a nonzero wavefunction and its Fourier transform cannot both vanish outside sets of finite measure. For $\hbar = 0$ with all states admissible, point states give $\alpha = \beta = 1$ for every $a, b > 0$, and there is no bound.

Proof. In the Schrödinger representation $p = -i\hbar\, d/dq$, so confining p to $[-b, b]$ is band-limiting the wavefunction to spatial frequencies $|\nu| \leq b/(2\pi|\hbar|)$. With the interval length $2a$, the Slepian parameter $c = \pi W T$ equals $\pi \cdot \frac{b}{2\pi|\hbar|} \cdot 2a = ab/|\hbar|$ (equivalently $c = \Omega T/2$ with the angular band $\Omega = b/|\hbar|$); the shifts q_0, p_0 are removed by a translation and a boost. With $D = P_a$, $B = Q_b$ one has $\|DB\| = \sqrt{\lambda_0(c)}$ (Slepian and Pollak 1961). For a pure state f: if $\alpha\beta = 0$ the left side is at least $\pi/2 > \arccos\sqrt{\lambda_0}$. Otherwise put $g = Df/\alpha$, $h = Bf/\beta$; then $\mathrm{Re}\langle f, g\rangle = \alpha$, $\mathrm{Re}\langle f, h\rangle = \beta$ and $|\langle g, h\rangle| = |\langle Df, DB\,Bf\rangle|/(\alpha\beta) \leq \sqrt{\lambda_0}$. The angle $d(x, y) = \arccos \mathrm{Re}\langle x, y\rangle$ is the geodesic distance on the unit sphere of the underlying real Hilbert space, so $\arccos\sqrt{\lambda_0} \leq d(g, h) \leq d(g, f) + d(f, h) = \arccos\alpha + \arccos\beta$ (proof supplied by the reviewer). Landau and Pollak (1961) (passage, as read by the reviewer in the archive.org scan) show that the inequality is sharp and characterize the attainable pairs in four cases; it binds only when $\alpha^2, \beta^2 \geq \lambda_0$ and holds everywhere as a necessary condition, which is all that is used here. Mixed states: in the coordinates $u = \alpha^2 + \beta^2 - 1$, $v = \alpha^2 - \beta^2$ the boundary is an arc of the ellipse $(u/\cos\theta_0)^2 + (v/\sin\theta_0)^2 = 1$, $\theta_0 = \arccos\sqrt{\lambda_0}$, centred at $(\frac{1}{2}, \frac{1}{2})$, inscribed in the unit square and tangent to its sides at $(\lambda_0, 1)$ and $(1, \lambda_0)$, where the arc ends. The allowed set is the square with that one corner rounded off, a convex set, and $(\langle P_a\rangle, \langle Q_b\rangle)$ is linear in the state. With $\alpha, \beta \geq \sqrt{1-\epsilon}$, $\arccos\sqrt{1-\epsilon} = \arcsin\sqrt{\epsilon}$ and $\cos(2\arcsin\sqrt{\epsilon}) = 1 - 2\epsilon$ give the second display ($\epsilon \leq \frac{1}{2}$ keeps the angles in the monotone range). For interval windows $\epsilon = 0$ is excluded because

$\lambda_0(c) < 1$; for general sets of finite measure it is the theorem of Amrein and Berthier (1977) and Benedicks (1985) (metadata). For $\hbar = 0$ the state $\delta_{(q_0,p_0)}$ lies in every window. □

Theorem E complements Theorem C. Theorem C bounds the disturbance that a record leaves; Theorem E bounds the concentration of the body's state that records can produce, which is the quantity Theorem I of the unit-and-indeterminacy note shows can be made arbitrarily small classically. It uses no Gaussian assumption, and it depends on the windows only through the product of half-widths ab in units of $\hbar$, the action scale of the comparison Newton takes to zero (the windows' area is $4ab$). It shares with the Planck paper's floor only the total-variation input $(1 - 2\epsilon)$; for small ϵ, Slepian's asymptotics give $c_*(\epsilon) \approx \frac{1}{2}\ln(1/\epsilon)$ (reviewer's recollection), a few units of $\hbar$ at practical error levels. (Check, 2026-09-27: the large-c formula $1 - \lambda_0(c) \simeq 4\sqrt{\pi c}\, e^{-2c}$ of Slepian 1965 and Fuchs 1964, recalled, printed passage not yet located, turns $\lambda_0 \geq (1-2\epsilon)^2 \simeq 1-4\epsilon$ into $2c - \frac{1}{2}\ln c - \frac{1}{2}\ln\pi \geq \ln(1/\epsilon)$, so $c_*(\epsilon) = \frac{1}{2}\ln(1/\epsilon) + \frac{1}{4}\ln\ln(1/\epsilon) + O(1)$; the leading term agrees with the reviewer.) It holds for every $\hbar \neq 0$; the state route of the routes note must reproduce it separately.

7. The same structure in Rivero 1998: the classical Dirac measure and its constant

The thesis that the path-integral constant is a consistency condition was stated by the user in 1998 (Rivero, "A short derivation of Feynman formula", arXiv:quant-ph/9803035, full read). Its abstract: "The complex exponential weighting of Feynman formalism is seen to happen at the classical level. (Finiteness of) Feynman path integral formula is suspected then to appear as a consistency condition for the existence of certain Dirac measures over functional spaces." Theorems A–D fit it point by point.

- **The complex exponential is classical.** The Dirac measure concentrated on the critical points of f, which is the principle of virtual work, is given the representation $\langle \delta(f') | g \rangle = \lim_{\varepsilon \to 0} \iint e^{i(f(y)-f(x))/\varepsilon} g(x)\, dx\, dy/\varepsilon$ (its eqs. (1)–(2)). The "halved" functional $\varepsilon^{-1/2} \int e^{if/\varepsilon} O\, dx$ recovers it by modulus squared, $g = |O|^2$, when f has a single critical point (eqs. (3)–(4); the paper proposes eq. (2) as asymptotically equivalent to eq. (1), with the normalization of the Dirac measure absorbing a factor 2π). The classical variational problem already carries an amplitude whose square is the record.
- **Theorem A is its Ehrenfest remark.** The paper's control transformation $\tau_\mu \delta^h = \delta^{\mu h}$ (eq. (10)) leaves the mean of the Euler–Lagrange expression invariant in a formal manipulation (the paper's word), "RG invariance in this context relates to Ehrenfest theorem" (eq. (11)): the equations of motion hold for every value of the constant. Theorem A(a) is the operator form.
- **The dilation family.** The map $h \mapsto \mu h$ is the isomorphism $*_{\lambda^2\hbar} \to *_\hbar$ of §4: every nonzero value of the constant gives an equivalent theory, and only the value 0 is distinguished.
- **Why Gaussian integrals are exact here.** For a quadratic f, the halved functional gives the classical value of every linear observable for every ε, because the odd moments of $e^{ia(x-x_*)^2/\varepsilon}$ about x_* vanish. The Gaussian integrals that run through this programme's notes (the elimination law of the refinement note, heat kernels, Gaussian marks, Astra's Gaussian restriction) are exact because Galileo's comparison and Newton's other integrable cases are quadratic; this is Theorem A(b) in the language of the Dirac measure. For general forces the object is the oscillatory measure.
- **Where a fixed constant enters.** On paths, the regularized measure has two parameters, the action resolution ε of the Dirac measure and the time step ε' of the partition (eq. (5)). Newton's classical mechanics takes $\varepsilon \to 0$ at each partition (the classical elimination, Proposition 2 of the refinement note) and then refines; the path integral of eqs. (7)–(8) holds the action resolution at a fixed h while the partition is refined. (The paper also writes $\varepsilon = h\varepsilon'$ after eq. (5), where the resolution itself goes to zero; the reading here follows eq.

(7).) For Newton's quadratic cases both orders converge (the refinement note's Proposition 2 and Theorem 3), which is the independence of Theorem A. The 1998 conjecture that finiteness forces the joint limit concerns the general, non-quadratic case, and it remains open.

- **The tangent groupoid and the ultimate velocity.** The paper proposes Connes's tangent groupoid as the frame (Rivero 1997, arXiv:dg-ga/9710026, abstract; Cariñena, Clemente-Gallardo, Follana, Gracia-Bondía, Rivero and Várilly 1999, abstract: the construction "generalizes the standard Moyal rule"). In that groupoid a pair of points (x, y) at scale $\varepsilon > 0$ is a chord, and the smooth structure glues it at $\varepsilon = 0$ to the tangent vector $\lim(y - x)/\varepsilon$. The $\varepsilon = 0$ boundary is Newton's *velocitas ultima*, the "certain and definite" limit of §1; functions there are functions of place and velocity together, a commutative algebra, while at $\varepsilon > 0$ the groupoid convolution of kernels is noncommutative. Joint determinacy holds on the boundary fibre, and the deformation of Theorem B is the passage to the interior with ε playing the part of $\hbar$. This is a reading of the cited constructions; on the Moyal side the boundary algebra is functions on T^*M, Fourier dual to TM.

8. What remains, and consequence for STATE

Theorems A–D are proved for one degree of freedom; several degrees follow with $Sp(2n)$, whose invariants of two vectors are again generated by the symplectic pairing. The analogy with the parallel postulate holds for the algebraic route under (H1)–(H3): independence from the Heisenberg-form axioms, a unique deformation with one constant, and a floor. It is partial in two ways stated above: (H2) is an added premise, and the state route gives the same floor without noncommutativity. The identification with Newton rests on his *velocitas ultima*, with the step from geometry to records interpretive.

For STATE item 2 this places the necessity question precisely. Theorem U supplies the value of the unit from radiation thermodynamics; Theorem I and the routes note say what must be denied; this note shows that the algebraic denial is a single step with a single constant, and that Newton asserted, in the ultimate velocity, the joint determinacy it denies.